Parts to buy

ModalAI Starling 2 Max — LED strip, cooling fan, propellers. Each item has its own page with a picture, exact specs and a box to write prices in.
Keep every receipt — ask for an itemised invoice showing the shop's name and GST registration number.

✓	Item	Key spec	Where
☐	**Addressable LED strip**	**WS2812B or SK6812** 5 V · 60 LED/m · need 16 pixels (≈27 cm) · buy 1 m (1 m also covers 6 px per arm if the arms re-measure at ~100 mm)	Sim Lim **Tower**
☐	**Cooling fan**	**25 × 25 × 7 mm · 5 V · 2 WIRES ONLY** ≤ 150 mA	Sim Lim **Tower**
☐	**JST SH 1.0 mm connector**	2-pin, with pigtail wires (×2–3 spare)	Sim Lim **Tower**
☐	**Resistors**	330 Ω or 470 Ω, 1/4 W (×10)	Sim Lim **Tower**
☐	**Hook-up wire**	Thin stranded, 3 colours, ~2 m each	Sim Lim **Tower**
☐	**Capacitor**	470 µF electrolytic, ≥ 6.3 V (×2) · goes at the strip end	Sim Lim **Tower**
☐	**Kapton tape** or clear heat-shrink	~10 mm tape, or heat-shrink to fit a 10 mm arm	Sim Lim **Tower**
☐	**Propellers**	**7 inch · 3.7 pitch · 3 blades** (7037-3 / 7×3.7×3)	Sim Lim **Square**
☐	*Backup kit* (buy all three or none)	5 V BEC ≥ 1 A · **74HCT125** or 74AHCT125 chip (×2) · XT30 Y-lead	Sim Lim **Tower**

Two buildings, side by side

Sim Lim TOWER

10 Jalan Besar
Electronic components
LED strip · fan · connectors · resistors · wire
Closes ~7:30 pm.

Sim Lim SQUARE

1 Rochor Canal Road
Drone / FPV parts
Propellers — ask for **Hobby Square**
Mon–Sat 12:30–7:30 pm.

1 · LED strip

addressable / NeoPixel type



The right type — flexible addressable strip



Count the wires: THREE

Photos: Cytron Technologies, product DS-6812-WS-601M. Example shown is 60 LED/m — exactly the density you want.

WS2812B or SK6812

5 V · **60 LED / metre** · **1 metre** · must be **individually addressable**

Width: **as narrow as possible** — arm is ~10 mm, thinner at the motor. **5 mm "narrow" strip ideal**, 10 mm bare strip is the maximum.

DO NOT BUY	Why
APA102 / DotStar / **SK9822** / **WS2801**	need a 4th clock wire — will not work
WS2815	12 V — wrong voltage
Non-addressable RGB strip	whole strip one colour only
Silicone-sleeved / waterproof tube strip (~14.5 mm)	wider than the ~10 mm arm

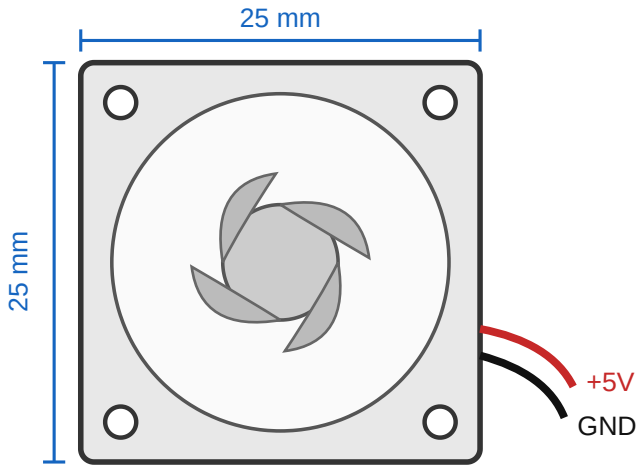
Ask the shop: "Is this individually addressable, 5 volt, with **3 wires** — data, +5 V, ground?" If it needs **4 wires**, it is the wrong type.

Quotes

Shop / unit no.	Price S$	Notes

2 · Cooling fan

size code “2507”



25 × 25 mm, 7 mm thick, exactly two wires

Drawn rather than photographed on purpose: the size and the wire count are what matter, and a photo of a different fan would mislead.

> **25 × 25 × 7 mm**
> **5 V DC** · **2 WIRES ONLY** · ≤ 150 mA
> Size code: **“2507”**

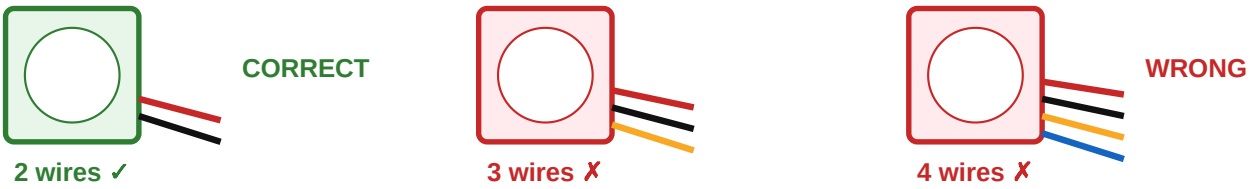
*Shops know this size as **2507**. Asking for “a 2507 5 volt 2-wire fan” is the fastest way to be understood.*

DO NOT BUY ✗	
12 V fan	wrong voltage
3-wire (has tacho)	will not work
4-wire (PWM input)	will not work

*Prefer a **ball bearing** fan over sleeve — it is going on a flying drone and will be vibrating.*

*30 mm or 20 mm is acceptable **only** if 25 mm is unavailable — ask me before buying a different size.*

Why the wire count matters



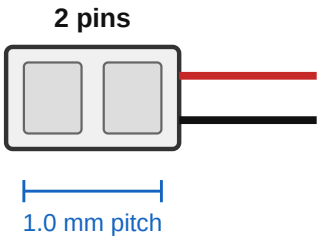
*The drone switches the fan's **ground** on and off to control speed. A 3- or 4-wire fan has extra pins that need a constant ground, so it will stutter or not start.*

Quotes — write the price here

Shop / unit no.	Exact item seen	Price S$	Bought?

3 · Connector, resistors, wire

small parts





orange-orange-brown = 330 Ω



This shape — 2-pin housing with two wires already crimped on

Photo: Christopher.nunez, CC BY-SA 3.0, via Wikimedia Commons. Shown for shape only — confirm the pitch is 1.0 mm.

JST SH 1.0 mm

2-pin, female, **with wires attached** (a “pigtail”)

*This is the **small** JST — 1.0 mm between pins. Do **not** buy JST PH (2.0 mm) or XH (2.5 mm); they are bigger and will not fit.*
Buy 2–3 spares, they are easy to damage.

330 Ω or 470 Ω

1/4 watt, ordinary carbon film
Buy **10** — they cost cents

Either value works. Anything between 300 and 500 Ω is fine.

Hook-up wire

Thin stranded, **3 colours** (red / black / any)
~2 m of each · 24–28 AWG

Capacitor · Kapton tape

470 µF electrolytic, ≥ 6.3 V (×2), across + and GND at the strip start; the stripe marks the negative leg.
~10 mm Kapton tape or heat-shrink that fits over a 10 mm arm, to hold the strip ends on the carbon.

Quotes — write the price here

Shop / unit no.	Exact item seen	Price S$	Bought?

3b · 5 V backup kit

all three, or none

Only needed if the strip misbehaves on the ESC’s 3.3 V pad. Cheap, so buying it now saves a second trip.

5 V BEC + 74HCT125 + XT30 Y-lead

The ESC's LED pad gives only **3.3 V**. If the strip is too dim or glitches on 3.3 V, power it from 5 V instead:
5 V BEC, 1 A or more, input **7–17 V** (runs from the 4S battery) · **XT30 Y-lead** to tap the battery for it · **74HCT125** or **74AHCT125** chip (×2) to lift the 3.3 V data signal to 5 V.

Do not buy 74HC125 or 74LS125 *— wrong input levels. A 5 V strip on 3.3 V data without this chip may work on the bench and fail in flight.*

Quotes — write the price here

Shop / unit no.	Exact item seen	Price S$	Bought?

4 · Propellers

7 inch tri-blade



Three blades, M5 centre bore



The bag to look for — label reads **GEMFAN 7037-3**

Photos: OddityRC / Gemfan 7037-3 product listing

7 × 3.7 × 3

7 inch diameter (178 mm) · **3.7** pitch · **3 blades** · **M5** bore
Ask for **“7037”**. Buy **2 sets of 4** (each set is 2 CW + 2 CCW).

PITCH MUST NOT EXCEED 3.7. Do not accept 7040, 7042, 7045 or any “freestyle” propeller — a higher-pitch prop will overheat the motors and ESC on this aircraft. 7035 (3.5 pitch) is an acceptable substitute.

The same propeller fits both of our aircraft, even though their frames and motors differ.

Quotes

Shop / unit no.	Price S$	Notes

Where to go

call before travelling

Components — Sim Lim Tower, 10 Jalan Besar

Shop	Known for	Buy here
J.C.W. Electronic	LED components, **DC fans**, power adapters	LED strip + fan — try here first, it may have both
Continental Electronic Enterprise	General hobby electronics	Resistors, wire, connectors
Kuriosity	Arduino / Raspberry Pi	Addressable LED strip, JST pigtails

Directory is ~2 years old — unit numbers may have changed. Most shops shut by ~7:30 pm regardless of posted hours. Prices vary between units; ask two or three.

Propellers — Sim Lim Square, 1 Rochor Canal Road

Hobby Square	FPV drone shop inside Sim Lim Square. Lists propellers from 1" to 12" including carbon fibre. Listed hours Mon–Sat 12:30 pm – 7:30 pm.

If Hobby Square has no 7037

Shop	Notes
Sky Hobbies skyhobbies.com.sg	Jalan Sultan area. Claims to be the largest HQProps stockist in Singapore.
Helicopter Smartfly smartfly.com.sg	Queensway Shopping Centre. Mon–Sat 12–7 pm, Sun 3–7 pm, **closed Wednesday**.
Rotor Hobby shop.rotor.com.sg	French Road, Kitchener Complex. Tel +65 6296 3686.

Fan fallback

Amicus amicus.com.sg	8 Ubi Road 2, #03-13 Zervex. Tel +65 6980 6783. Stocks 5 V brushless fans with order collection — **call first**, the listed model is 60 mm not 25 mm.